Round 3 Student Challenge Brief

Quality Incident and Action Assistant

Your final challenge

Your manufacturing portal can search manuals and inspect product images. Connect these capabilities so an Engineer can turn a saved Defective inspection into a documented quality incident. Retrieve the applicable quality SOPs, generate a cited summary and suggested actions, and obtain a Supervisor review.

Item	Requirement
Participants	Finalist teams; 3 or 4 students per team
Development	1.5 Hour
Submission deadline	1:15 PM on Day 2
Evaluation	Day 2, 1:15 PM to 2:30 PM
Final ranking	Round 3 score out of 100 determines the winners

Keep the build focused

Reuse login, role permissions, saved inspections, the model, RAG, database connections, and UI from earlier rounds. Add one incident screen, a few endpoints, and a small database extension. No new model, training, image labeling, or factory integration is required. Synchronous SOP indexing and generation are acceptable.

Expected workflow

- Engineer selects their own completed Round 2 inspection and requests an incident.
- For an original Defective prediction, create one incident linked to that inspection. Repeated requests return the existing incident.
- Use the product ID to select the mapped SOPs, retrieve relevant passages, and generate a short summary and suggested next steps with verifiable citations.
- Supervisor reviews the evidence and records Approved for follow-up or Needs more information with a required comment.
- Reload the incident to show the saved inspection, original prediction, generated text, sources, and review.

Input pack

SOPs contains three short text-based PDFs. product_sop_mapping.csv identifies the applicable documents and revisions. P-PLATE01 is the same product used in Round 2. The P-GASKET02 material is an unrelated reference for source-isolation checks; you do not need to build another inspection model or register that product.

Rules and Acceptance Requirements

Inspection eligibility and duplicate handling

Use the immutable original model prediction from Round 2 for this round's eligibility rule. Only completed Defective inspections may create incidents. Display any human correction separately; it does not change eligibility in this challenge. Normal, queued, processing, or failed inspections must return a clear message without creating an incident.

Enforce one incident per inspection in persistent storage, including repeated or simultaneous requests. An existing incident must retain its ID and any Supervisor review. Repeated requests must not create a fresh record or silently overwrite saved recommendations.

Product scope and grounded output

Match product_id exactly against product_sop_mapping.csv. Restrict retrieval to the mapped document IDs and revisions before generation. Both P-PLATE01 SOPs are complementary required references for the complete workflow. Never use a different product's SOP or the Round 1 equipment manuals as quality guidance.

Generate a factual incident summary from the saved inspection and a short list of suggested actions from retrieved SOP passages. Every suggested action needs a source document name or ID, revision, one-based PDF page, supporting passage, and source link. The original prediction is not proof of the defect subtype, cause, or severity.

Missing evidence and review decisions

If the mapping is absent, a required SOP is missing or unreadable, retrieval yields no relevant passages, or generation fails, keep one incident with evidence status Insufficient and a readable reason. Show Manual review required and omit unsupported suggested actions. Do not use general LLM knowledge to fill the gap.

Use evidence status Sufficient only when the applicable evidence supports the generated actions. Store review status separately: Pending, Approved for follow-up, or Needs more information. Insufficient-evidence incidents cannot be approved; a Supervisor may record Needs more information with a comment. Approval authorizes follow-up only, not product release.

Access and persistence

Engineers create and view incidents only for their own inspections. Supervisors view all incidents and submit decisions. Administrator status alone does not grant review access. Enforce these rules on the server. Store the incident ID, inspection reference, product, original model result and version, summary, actions, evidence status and reason, source references, creator, timestamps, decision, reviewer, and comment.

Do not overwrite Round 2 predictions or reviews. Keep the two rounds' frozen submissions separate. Suggested checks cover duplicate requests, rejection of Normal results, missing evidence, and Engineer review denial; use your existing test framework.

Submission Demonstration and Scoring

Demonstration checklist

- Select a saved completed Defective inspection, create an incident, and open a cited SOP page. Show that the two plate SOPs support the workflow.
- Repeat the same request and show that the incident ID and record count remain unchanged.
- Select a Normal inspection and show that no incident is created.
- Using a fresh eligible inspection in your test setup, make a required SOP or mapping unavailable and show the manual-review path. Restore the setup afterward.
- Attempt review as Engineer and show denial. Review as Supervisor, add a comment, and reload the saved result.

Example:

Reload the reviewed incident. Show the inspection, original output, recommendations, sources and saved review. Explain which earlier-round capabilities were reused.

Expected outcome: The same incident and its linked records remain available, including evidence status/reason where relevant, citations and review comment. The team explains how saved CV inspections feed the SOP retrieval and generation workflow, followed by a separate human decision.